Simulation Arena

# Deal Approval & Margin Balancing

**Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $3,835.12M of expected profit.**

Seed 42 | Run run_d2f452505e70 | 170,000 scenarios tested | Monte Carlo simulation with an empirica… | 22 Sep 2026, 05:06

Same seed, same numbers - this run can be reproduced exactly.

## How this works

For each discount level we ran thousands of 'what-if' worlds. In each one the buyer's odds of saying yes wobble a little, and your delivery cost wobbles a little. We then ask the only question that matters: if you offered this discount again and again, how much profit would you bank on average? That is the expected profit - the chance of winning multiplied by the profit you keep. Because those worlds are a sample, we also work out how much the answer itself would wobble if we ran it again. Any discount whose expected profit is within twice that wobble of the best - or within 1% of the best profit, whichever is wider - is treated as giving the same answer. That is the range where you have genuine room to negotiate.

## The numbers that matter

**BEST DISCOUNT TO OFFER**
**0%**
This is where the extra wins are still worth more than the margin given away.
vs rep asked for 25%

**EXPECTED PROFIT AT THAT DISCOUNT**
**$489.79M**
Average profit if you made this call many times over.

**DISCOUNT RANGE YOU CAN APPROVE WITHOUT LOSING MONEY**
**0%**
Only one discount stands out - the model cannot tell any neighbouring discount apart from it. The model's own margin of error here is about $38.00M, so treat this as the best single number - the next discount along already costs real money.
vs rep asked for 25% - outside the range

**WHERE IT STARTS TO GENUINELY COST YOU**
**2%**
At 2% the expected profit falls $191.78M below the best. That gap is wider than the model's own margin of error, so the loss is real.
vs costs about $191.78M

**EXPECTED PROFIT IF YOU CAVE IN**
**-$3,345.34M**
What 25% actually earns you.
vs -$3,835.12M vs the optimum

**PROFIT GIVEN AWAY**
**$3,835.12M**
Buying +49 points of win probability costs you this much.

**ODDS OF WINNING AT THE BEST DISCOUNT**
**36.9%**
The chance the deal lands.

**WORST CASE (1-IN-20)**
**$0**
In an unlucky run of 20 deals, this is roughly the profit you would see.

## How we got there

### 1. The two forces fighting each other

Every point of discount does two things: it makes the buyer a bit more likely to say yes, and it takes $209.12M straight off your profit. Early on the first effect wins. Later, the second one does.

Evidence: Cost to deliver $19,593.53M on $20,911.98M of list value

### 2. What the buyer's behaviour looks like

They start around 35.0% likely to buy at full price and never become certain - the curve flattens near 95.0%. Around 17% discount is where they are most persuadable.

Evidence: Curve source: 20000 of your past deals

**3. The sweet spot**

0% discount gives the highest expected profit at $489.79M, with a 36.9% chance of winning.

Evidence: peak of 17 levels tested

**4. The range where the answer is the same**

0% is the only discount that stands out; the model cannot tell any neighbouring discount apart from it. The next level along already costs $191.78M.

Evidence: margin of error $38.00M; safe range 0%

**5. What the requested discount really costs**

Moving from 0% to 25% buys +49 points of win probability but costs $3,835.12M of expected profit - more than the model's margin of error, so the loss is real.

Evidence: -$3,345.34M vs $489.79M

## Charts

### Where profit peaks

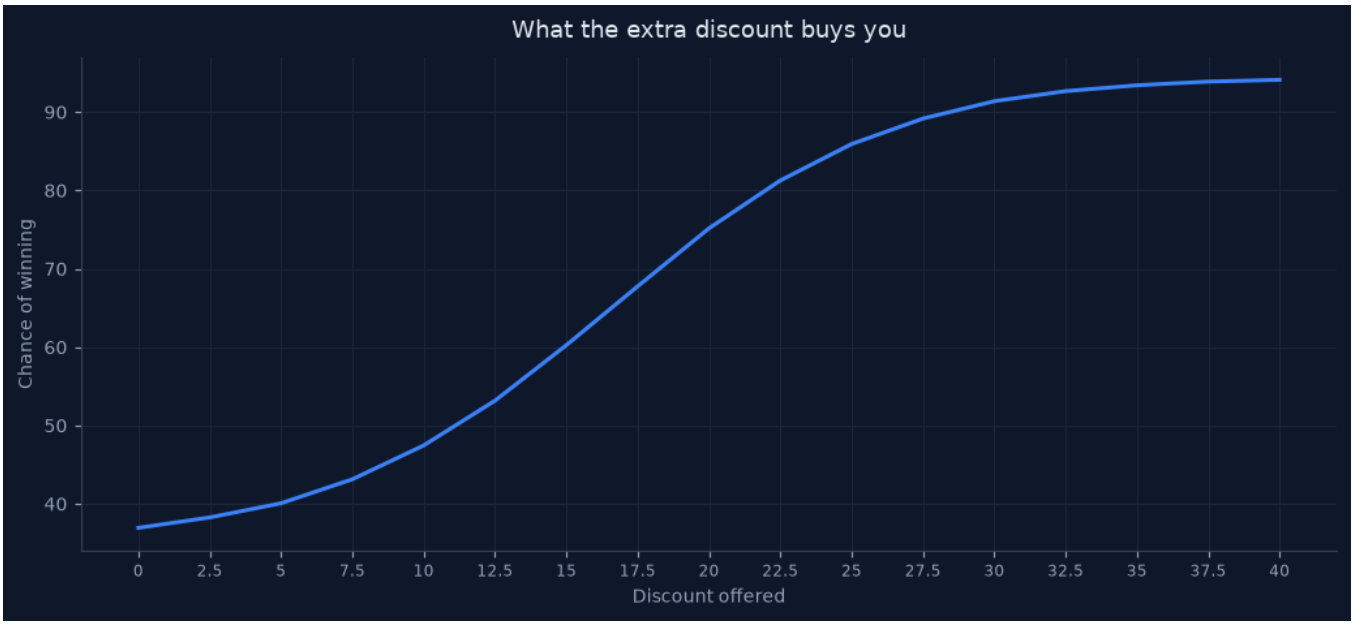
Everything from 0% gives the same answer within the model's margin of error; the peak is 0%




*What this shows: Grey is what you keep per won deal - it only falls. Green is what you actually expect to earn - it rises, peaks, then falls. The amber dashed line is the best profit minus the model's own margin of error: every discount whose green line stays above it is the same answer, which is the 0% range. Past that you are genuinely paying for the win.*

### What the extra discount buys you

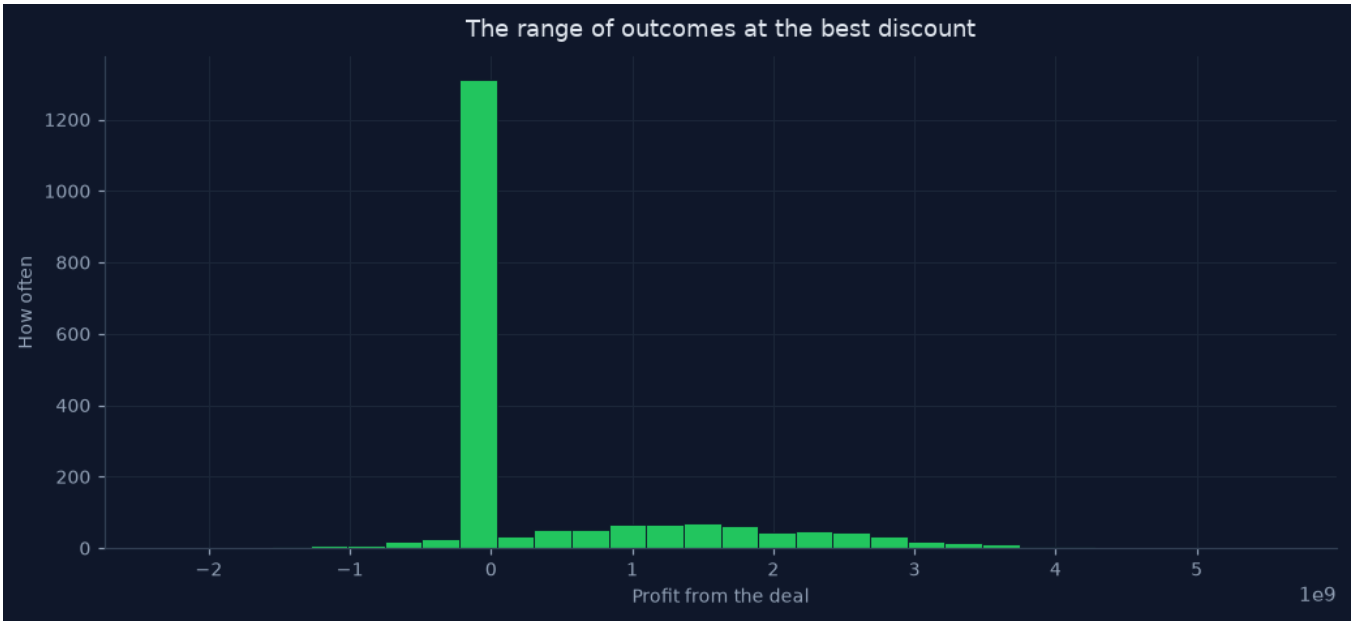
Curve learned from 20000 of your past deals




*What this shows: Past the knee of this curve, extra discount buys almost no extra certainty - but keeps costing margin. That is the trap.*

### The range of outcomes at the best discount

10,000 simulated worlds at 0% discount




*What this shows: The cluster at zero is the deals you lose. The spread on the right is the profit you make when you win.*

## Tables

### Every discount level tested

Expected profit = chance of winning x profit kept. Highlighted row is the winner.

| Discount | Price after discount | Profit if won | Win chance | Expected profit | Worst case | Breaches floor? |
|---|---|---|---|---|---|---|
| 0.0% | $20,911,979,419 | $1,316,300,252 | 36.9% | $489,788,040 | $0 | 85% chance |
| 2.5% | $20,389,179,933 | $806,594,813 | 38.3% | $298,003,747 | -$518,160,848 | 92% chance |
| 5.0% | $19,866,380,448 | $271,925,292 | 40.1% | $107,035,517 | -$1,106,043,708 | 96% chance |
| 7.5% | $19,343,580,962 | -$257,233,422 | 43.1% | -$108,229,562 | -$1,667,770,697 | 99% chance |
| 10.0% | $18,820,781,477 | -$777,711,373 | 47.5% | -$359,508,475 | -$2,164,567,940 | 100% chance |
| 12.5% | $18,297,981,991 | -$1,260,773,237 | 53.2% | -$665,850,777 | -$2,766,028,834 | 100% chance |

| Discount | Price after discount | Profit if won | Win chance | Expected profit | Worst case | Breaches floor? |
|---|---|---|---|---|---|---|
| 15.0% | $17,775,182,506 | -$1,811,949,781 | 60.3% | -$1,098,691,257 | -$3,453,761,503 | 100% chance |
| 17.5% | $17,252,383,021 | -$2,312,040,617 | 67.8% | -$1,562,362,933 | -$4,013,868,439 | 100% chance |
| 20.0% | $16,729,583,535 | -$2,849,981,803 | 75.2% | -$2,143,707,264 | -$4,636,785,211 | 100% chance |
| 22.5% | $16,206,784,050 | -$3,382,134,825 | 81.3% | -$2,759,835,270 | -$5,202,826,247 | 100% chance |
| 25.0% | $15,683,984,564 | -$3,911,588,991 | 85.9% | -$3,345,336,695 | -$5,754,713,952 | 100% chance |
| 27.5% | $15,161,185,079 | -$4,442,147,949 | 89.2% | -$3,950,806,161 | -$6,295,583,075 | 100% chance |
| 30.0% | $14,638,385,593 | -$4,982,764,419 | 91.4% | -$4,556,191,989 | -$6,898,065,080 | 100% chance |
| 32.5% | $14,115,586,108 | -$5,484,029,348 | 92.7% | -$5,082,629,649 | -$7,407,158,713 | 100% chance |
| 35.0% | $13,592,786,622 | -$6,019,753,538 | 93.4% | -$5,631,830,290 | -$7,901,217,889 | 100% chance |
| 37.5% | $13,069,987,137 | -$6,505,024,880 | 93.9% | -$6,104,155,757 | -$8,375,361,830 | 100% chance |
| 40.0% | $12,547,187,651 | -$7,047,045,555 | 94.1% | -$6,637,161,913 | -$8,937,063,215 | 100% chance |

Row 1 maximises expected profit.

### The decision in one line

What changes if you approve the discount being asked for.

| Column 1 | Best (0%) | Asked (25%) | Difference |
|---|---|---|---|
| Discount | 0.0% | 25.0% | +25.0 pts |
| Win chance | 36.9% | 85.9% | +49.0 pts |
| Profit if won | $1,316,300,252 | -$3,911,588,991 | -$5,227,889,243 |
| Expected profit | $489,788,040 | -$3,345,336,695 | -$3,835,124,735 |
| Range where the answer is the same | 0% | outside it | margin of error $38.00M |

### The range where the answer is the same

Every discount we tested, how far it falls short of the best, and whether that shortfall is real money or just the model's own margin of error.

| Discount | Expected profit | Short of the best | Inside the range? | What it means |
|---|---|---|---|---|
| 0.0% | $489,788,040 | $0 | Inside | safe to approve |
| 2.5% | $298,003,747 | $191,784,293 | Outside | costs you $191.78M |
| 5.0% | $107,035,517 | $382,752,523 | Outside | costs you $382.75M |
| 7.5% | -$108,229,562 | $598,017,602 | Outside | costs you $598.02M |
| 10.0% | -$359,508,475 | $849,296,515 | Outside | costs you $849.30M |
| 12.5% | -$665,850,777 | $1,155,638,816 | Outside | costs you $1,155.64M |
| 15.0% | -$1,098,691,257 | $1,588,479,296 | Outside | costs you $1,588.48M |
| 17.5% | -$1,562,362,933 | $2,052,150,972 | Outside | costs you $2,052.15M |
| 20.0% | -$2,143,707,264 | $2,633,495,304 | Outside | costs you $2,633.50M |
| 22.5% | -$2,759,835,270 | $3,249,623,310 | Outside | costs you $3,249.62M |
| 25.0% | -$3,345,336,695 | $3,835,124,735 | Outside | costs you $3,835.12M |
| 27.5% | -$3,950,806,161 | $4,440,594,201 | Outside | costs you $4,440.59M |
| 30.0% | -$4,556,191,989 | $5,045,980,028 | Outside | costs you $5,045.98M |
| 32.5% | -$5,082,629,649 | $5,572,417,689 | Outside | costs you $5,572.42M |
| 35.0% | -$5,631,830,290 | $6,121,618,330 | Outside | costs you $6,121.62M |
| 37.5% | -$6,104,155,757 | $6,593,943,796 | Outside | costs you $6,593.94M |
| 40.0% | -$6,637,161,913 | $7,126,949,953 | Outside | costs you $7,126.95M |

Highlighted rows are the range where the answer is the same (0%) - within about $38.00M of the best.

### Recommended discount per deal (top 25)

The same logic applied to each deal individually.

| Deal | Customer | Value | Margin | Best discount | Expected profit |
|---|---|---|---|---|---|
| Q-000011589 | Acct-226235 | $24,000,000 | -5.7% | 0% | -$502,859 |
| Q-000009396 | Acct-223884 | $24,000,000 | 3.8% | 0% | $337,013 |
| Q-000009047 | Acct-090640 | $24,000,000 | 4.8% | 0% | $424,814 |
| Q-000008091 | Acct-110274 | $19,296,863 | 15.4% | 0% | $1,096,718 |
| Q-000007533 | Acct-062342 | $16,052,463 | 0.7% | 0% | $40,337 |
| Q-000018826 | Acct-138872 | $14,565,883 | 23.4% | 0% | $1,260,595 |
| Q-000002411 | Acct-019732 | $14,340,308 | -11.8% | 0% | -$623,716 |
| Q-000006273 | Acct-125458 | $14,145,541 | 46.0% | 20% | $2,766,932 |
| Q-000005246 | Acct-233590 | $14,094,974 | 12.6% | 0% | $654,193 |
| Q-000013379 | Acct-205409 | $13,878,192 | 21.0% | 0% | $1,078,510 |
| Q-000004222 | Acct-219531 | $13,822,773 | 2.4% | 0% | $123,102 |
| Q-000009736 | Acct-113136 | $13,452,114 | -3.5% | 0% | -$172,493 |
| Q-000017300 | Acct-007220 | $13,422,614 | 25.1% | 0% | $1,245,477 |
| Q-000000139 | Acct-108061 | $13,011,806 | -15.3% | 0% | -$736,628 |
| Q-000009880 | Acct-201403 | $12,782,137 | -22.3% | 0% | -$1,053,320 |
| Q-000001755 | Acct-058430 | $12,706,362 | -8.4% | 0% | -$393,945 |
| Q-000001411 | Acct-122168 | $12,567,749 | 23.2% | 0% | $1,077,915 |
| Q-000010633 | Acct-176419 | $12,154,931 | 1.3% | 0% | $59,290 |
| Q-000013798 | Acct-106373 | $11,654,026 | 18.1% | 0% | $779,052 |
| Q-000010069 | Acct-070297 | $11,489,319 | 15.7% | 0% | $665,721 |
| Q-000013273 | Acct-130561 | $11,410,414 | 27.0% | 0% | $1,136,772 |
| Q-000007138 | Acct-039861 | $11,334,211 | 18.2% | 0% | $763,956 |
| Q-000005723 | Acct-178920 | $11,250,338 | 16.4% | 0% | $680,975 |
| Q-000001338 | Acct-100033 | $11,225,300 | 17.8% | 0% | $739,607 |
| Q-000009264 | Acct-188657 | $11,173,578 | 22.0% | 0% | $908,792 |

## What we recommend

**Approve 0% - it is the only discount the model can tell apart from the rest. The 25% the rep is asking for is past it - that costs about $3,835.12M of expected profit.**

**Risk High** 100 out of 100 | Confidence: High

### Why

• Only 0% stands out from the rest - the model cannot tell any neighbouring discount apart from it.

• Expected profit peaks at 0% ($489.79M).

• The requested 25% delivers -$3,345.34M - $3,835.12M less than the peak, which is more than the model's margin of error.

• Win probability only improves +49 points for that giveaway.

### What to do

**1.** Approve 0% as the standard concession - it is the only discount the model can tell apart from the rest.

**2.** Above 0% (the top of the safe range), require VP sign-off: the 25% being asked for costs about $153.40M per point of discount.

**3.** At 25% there is a 100% chance of breaching the 12.0% margin floor - add a volume or term commitment to earn the deeper discount.

**The trade-off:** You are trading $3,835.12M of expected profit for +49 points of win probability.

## What we assumed and what to watch

### Worth knowing before you act

- At 25% discount there is a 100% chance the deal actually costs you money to deliver even if you win it.

### Assumptions

- List value $20,911,979,419; cost to deliver $18,937,406,936 at the 9.4% standard margin your file implies, plus $656,121,405 of consumption effects on top, giving a total cost base of $19,593,528,342 and an effective 6.3% margin once those effects are included.
- Consumption effects: 0.2% of value is consumed as unmetered burst you still pay for; 4 months of rollover defers revenue.
- Win probability curve: from 35.0% at full price toward a ceiling of 95.0%, halfway at 16.6% discount (source: 20000 of your past deals).
- Win probability uncertainty: plus or minus 5 points.
- Cost volatility: plus or minus 6% of cost.
- Margin floor: 12.0%.
- Two discounts are treated as giving the same answer when the gap between their expected profits is smaller than twice the model's own margin of error at both discounts added together (about $38.00M at the best one), or smaller than 1% of the best profit - whichever is larger.
- The range where the answer is the same is 0%, taken from the 17 discount levels we tested.
- 10,000 simulations per discount level; seed 42 (re-running gives identical numbers).
- Deal / quote ID -> your 'deal_id' column
- Customer name -> your 'customer' column
- Deal value (list price) -> your 'deal_value' column
- Cost to deliver -> your 'cost' column
- Standard margin (%) -> your 'margin_pct' column
- Discount given (%) -> your 'discount_pct' column
- Won? (1/0) -> your 'won' column
- Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column
- Unmetered burst used -> your 'burst_allowance_pct' column
- Rollover window -> your 'rollover_window_months' column

## Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

| Time | Stage | What happened |
|---|---|---|
| 0.00s | prepare | Won? (1/0) -> your 'won' column |
| 0.00s | prepare | Hit a consumption tier accelerator -> your 'tier_accelerator_hit' column |
| 0.00s | prepare | Unmetered burst used -> your 'burst_allowance_pct' column |
| 0.00s | prepare | Rollover window -> your 'rollover_window_months' column |
| 0.57s | prepare | Read 20,000 records into the model. |
| 0.57s | simulate | Running the simulation… |
| 0.57s | read | Reading the deal you gave us... |
| 0.57s | read | Found 20000 deal(s) worth $20,911.98M at an average 9.4% margin. |

| Time | Stage | What happened |
|---|---|---|
| 0.57s | prepare | Cost to deliver is about $18,937.41M. |
| 0.58s | prepare | Added the consumption effects from your file: $656.12M of extra cost to serve, taking the cost base to $19,593.53M. |
| 0.58s | prepare | Working out how much a discount really moves the buyer... |
| 0.85s | prepare | Learned the buyer's behaviour from 20000 past deals (they flip hardest around 16.6% discount). |
| 0.85s | simulate | Testing 17 discount levels, 10,000 outcomes each (170,000 scenarios in total)... |
| 0.85s | simulate | Simulated 11% of the discount levels... |
| 0.85s | simulate | Simulated 23% of the discount levels... |
| 0.85s | simulate | Simulated 35% of the discount levels... |
| 0.85s | simulate | Simulated 47% of the discount levels... |
| 0.85s | simulate | Simulated 58% of the discount levels... |
| 0.85s | simulate | Simulated 70% of the discount levels... |
| 0.85s | simulate | Simulated 82% of the discount levels... |
| 0.86s | simulate | Simulated 94% of the discount levels... |
| 0.86s | simulate | Simulated 100% of the discount levels... |
| 0.86s | summarize | All discount levels tested. Looking for the sweet spot... |
| 0.86s | summarize | Checked how much the answer would wobble if we ran it again - the range where the answer is the same is 0%. |
| 0.87s | summarize | Done. Best discount is 0% ($489.79M expected profit); the range where the answer is the same is 0%. |